

Paper OOL-7: Aromatic Stabilization, Chemical Alphabets, and Homochirality: Pattern Survival Before Genetic Code

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Abstract

This paper separates durable chemical alphabets and homochirality from autocatalytic closure. Aromatic scaffolds, redox cofactors, nucleobase-like motifs, and disordered aromatic polymers are treated as persistence mechanisms that may preserve patterns under flux noise. Homochirality is kept chemical and testable: CDFD requires amplification mechanisms, cross-inhibition, solubility effects, or template bias rather than unsupported historical certainty. The companion script separates aromatic persistence from a model-space chirality attractor.

Symbol Conventions

CDFD origins-of-life symbol convention.

Symbol	Meaning in Part II	Origins-of-life reading
Φ	Available driving flux	Redox, thermal, photochemical, proton/ion, or chemical potential drive supplied by the environment
C	Effective constraint or resistance	Mineral geometry, pore confinement, diffusion barrier, permeability, activation barrier, or maintenance burden
S	Responsive organization	Surface, boundary, catalytic network, coacervate, or protocellular structure that routes flux into retained function
M_s	Retained structural memory	Composition, sequence, topology, compartment state, mineral imprint, or boundary history that persists across renewal cycles
Ψ_s	Local adaptive ratio $(\Phi/C) \cdot S \cdot M_s$	Bookkeeping gauge for whether drive, constraint, responsiveness, and retention are co-present; not a universal constant
Λ_{life}	Integrated Life Number where used	Capstone surplus ratio for decay, near-critical proto-biology, and sustained self-maintenance

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Defines how fast a constraint, pathway, or retained structure grows under load
β	Relaxation, decay, or washout coefficient	Defines recovery, loss, clearance, hydrolysis, or return toward baseline
γ	Coupling, diffusion, or redistribution coefficient	Defines spread across pores, surfaces, compartments, networks, or chemical phases
δ, ϵ	Perturbation, tolerance, or small offset scales	Used only as local thresholds, numerical guards, or experimental tolerances
η, λ, μ, ν	Efficiency, rate, baseline, or frequency terms	Meaning is set by the equation and subscript where they occur
ρ, σ, τ, ϕ	Density/correlation, conductance/noise, time-scale, or phase/field terms	Never universal constants unless a paper explicitly defines them that way
Ω, Λ	Domain/state space; Life Number where used	Ω names a modeled state space; Λ is reserved for life-number notation only where defined

1 Introduction

Paper 6 addressed closure of reaction networks. Even with closure, most polymers hydrolyze or scramble without structural motifs that resist perturbation. Aromatic rings provide deep energy wells via delocalization (Hückel $4n+2$ rule in simplified form) and participate in electron–proton coupled chemistry [7].

Gap from Paper 6: closure without durable patterns cannot carry information.

Status: Inferred

2 Possible Sources of Aromatic Feedstock

The CDFD claim is source-agnostic. Aromatic feedstock could be produced locally by terrestrial photoredox, cyanosulfidic, mineral-surface, or thermal chemistry, and it could also be supplemented by exogenous delivery. Carbonaceous meteorites, especially Murchison, contain extraterrestrial organic matter and reported aromatic hydrocarbons [8, 13, 11, 15]. This does not prove that the first useful aromatics on early Earth arrived from space. It widens the reservoir of molecules that a surface, pore, film, or cycling environment could retain and test.

In the notation of the series, provenance is upstream of function:

$$A_{\text{eff}} = \Xi_{\text{loc}} (A_{\text{terr}} + A_{\text{exo}}), \quad (1)$$

where A_{terr} is terrestrial aromatic production, A_{exo} is externally delivered aromatic feedstock, and Ξ_{loc} is the retention/localization factor. A meteoritic pulse that disperses without retention does not become information-relevant. A smaller mixed pool that is trapped, cycled, and coupled to redox chemistry can matter more.

Status: Inferred

3 Functional Role of Aromatics

Aromatics have two different roles that must not be collapsed. Ordered π -systems can participate in energy capture and electron transfer; disordered aromatic or polymeric systems can absorb radicals, smooth redox fluctuations, and dissipate excess energy. Neither role requires literal chlorophyll or literal melanin at the origin of life. This paper focuses on pattern persistence: how aromatic scaffolds can survive flux noise long enough to become information-relevant.

Paper 11 uses B_{stab} for the total stabilization burden in $J = I_E \sigma_e \sigma_p / B_{\text{stab}}$. Here we use $\mathcal{K}_{\text{pat}}$ for pattern resilience under degradation noise, keeping molecular persistence separate from whole-system overload management.

Status: Inferred

4 Model and Assumptions

4.1 Electronic stabilization

Delocalized π orbitals lower energy relative to localized alternatives,

$$E_{\text{aromatic}} < E_{\text{localized}}. \quad (2)$$

Quinones illustrate reversible redox shuttles,



Status: Inferred

4.2 Flux coupling

Reaction rates for aromatic redox couples scale with available electron and proton currents,

$$R_Q \propto |\mathbf{J}_e \cdot \mathbf{J}_p|, \quad (4)$$

in the same phenomenology used throughout the series.

Status: Hypothesis

4.3 Pattern information (scalar proxy)

Let P be pattern survivability. A crude Shannon-like proxy is $I \sim -\log P$. Under flux noise, aromatic motifs maintain larger P than flexible aliphatic chains in the simulation.

Status: Hypothesis

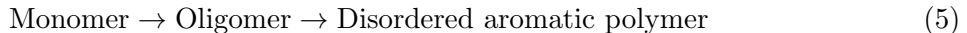
4.4 Notation guard

Draft manuscripts sometimes wrote $S \propto 1/\Lambda$ for pattern stability. We rename this to $\mathcal{K}_{\text{pat}}$ to avoid confusion with tri-regime stabilization burden B_{stab} in $J = I_E \sigma_e \sigma_p / B_{\text{stab}}$ (Paper 11).

Status: Derived

5 Disordered Aromatic Polymers as Pattern Stabilizers

As aromatic molecules accumulate and interact, polymerization becomes possible through oxidative coupling reactions. One possible outcome is a heterogeneous, disordered aromatic polymer. In this paper the claim is deliberately narrow: polymeric aromatic networks can improve persistence of molecular patterns under flux noise. The broader overload-buffering role, including modern melanin as an example, is deferred to Paper 11 after photochemical energy capture has been introduced.



5.1 Electronic Structure

Unlike simple aromatics, disordered aromatic polymers can exhibit extended π -electron delocalization, structural heterogeneity, and a broad density of electronic states. This creates a quasi-continuous band structure:

$$E(k) \rightarrow \text{continuous distribution} \quad (6)$$

5.2 Broadband Absorption and Redox Capacity

Such polymers can absorb across broad spectral bands,

$$J_{\text{abs}} = \int \sigma(\lambda) I(\lambda) d\lambda, \quad (7)$$

and may reversibly store charge:



This allows for electron buffering and smoothing of redox gradients in model space. At this stage of the series the point is pattern resilience, not yet whole-system photoprotection.

5.3 Role in Information Stability

Disordered aromatic stabilizers can protect information-bearing scaffolds by reducing chemical noise and stabilizing local electron flow. Thus, stable information I_{stable} can be represented schematically as:

$$I_{\text{stable}} \propto \frac{J_e J_H}{C_{\text{eff}}} \quad (9)$$

with $C_{\text{eff}} = C + C_{\text{buffer}}$.

5.4 Interpretation

The important transition is from single-molecule persistence to a matrix capable of preserving patterns. Paper 11 then asks what happens when external input increases enough that stabilization must become a system-level overload function.

Status: Inferred

6 Main Results

6.1 Conceptual

Aromatics bridge metabolism and proto-genetics: quinones shuttle energy; nucleobase-like aromatics support hydrogen-bond patterning [5]. Stacking (π - π) organizes spatial order.

Status: Inferred

7 Numerical Verification

Companion script: `supplementary_o01_07.py`.

- Final aliphatic pattern strength: 0.018
- Final aromatic pattern strength: 1.000
- Chirality trajectory (model coordinates): $(L, R) = (1.011, 0.989) \rightarrow (23.818, 0.007)$
- Best retained mixed-source functional score: 0.610

Large separation between aromatic and aliphatic endpoints supports differential persistence *under the scripted degradation law*. Chirality divergence is a *model-space attractor*; no claim is made that terrestrial biomolecular handedness follows this mechanism without laboratory tests. The source-mix diagnostic writes `aromatic_source_mix.csv`; it is a provenance guardrail, not historical proof. It shows that exogenous supply helps only when feedstock is retained and functionally coupled.

Status: Derived

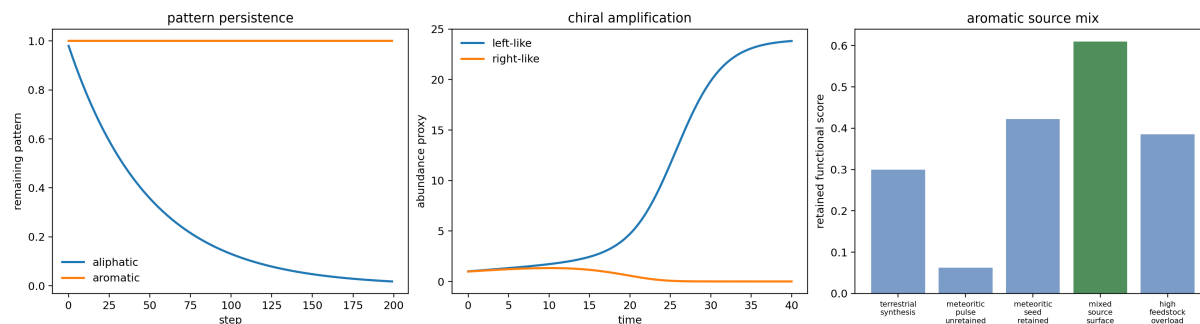


Figure 1: Merged aromatic-information diagnostic. The pattern panel shows differential survival under the scripted degradation law. The chirality panel is a model-coordinate divergence, not a historical claim about terrestrial handedness. The source-mix panel separates aromatic provenance from functional selection: delivery only matters when retained and coupled.

8 Interpretation

The simulation illustrates why aromatic scaffolds are privileged in prebiotic reasoning: they couple redox chemistry to structural persistence [1]. Real selectivity requires stereospecific catalysts and racemization kinetics absent here.

Status: Inferred

9 Chiral Amplification Without Overclaiming

Earlier drafts allowed chiral bias to drift into speculative physics. The release version is stricter. Part I permits CDFD to discuss chirality as a constraint-history variable [9], but Part II must anchor biological handedness in chemistry. Recent work on prebiotic ligation shows that symmetry breaking and chiral amplification can arise in competitive peptide-forming reactions under plausible chemical constraints [2]. The CDFD claim is therefore not that vacuum chirality chooses terrestrial handedness. The claim is that once a small imbalance exists, autocatalytic or ligation networks can amplify it if cross-inhibition, solubility, templating, or differential persistence make one handedness less lossy.

Status: Hypothesis/Inferred

10 Chemical Alphabets Before Genetic Code

Chemical memory requires a bridge from persistent structure to alphabet-like distinctions. CDFD does not claim that nucleobases appear fully formed at this stage. The stronger formulation is that several molecular classes can provide primitive alphabets:

- redox cofactors and quinone-like species encode electron-state differences;
- short peptides or amino-acid-like oligomers encode catalytic surface preferences;
- nucleobase-like aromatics encode hydrogen-bonding and stacking patterns;
- disordered aromatic polymers encode persistence and redox-buffer states, but not sequence information by themselves.

The transition to heredity requires that at least one alphabet becomes templatable, copyable, and selectable. Paper 9 takes up that stricter requirement.

Status: Inferred

11 Wet–Dry and Cycling Environments

Autocatalytic closure may need more than one steady environment. Wet–dry cycles, freezing/thawing, tidal concentration, evaporation, and vent pulsation can alternately concentrate monomers, drive condensation, rehydrate products, and select stable networks. In the present framework these cycles modulate both $\Phi(t)$ and $C(t)$:

$$\Gamma_{\text{closure}}(t) = \frac{k_{\text{auto}} c(t) L(t)^2 C(t)}{1 + \ell_{\text{side}}(t)}. \quad (10)$$

The relevant question is whether a network survives the full cycle, not whether it grows in a single favorable interval.

Status: Hypothesis

12 Cyanosulfidic and Mixed-Precursor Chemistry

Metabolism, peptides, lipids, and nucleotides should not be treated as isolated chemical universes. The cyanosulfidic chemistry program demonstrates that RNA, amino-acid, and lipid precursor chemistry can be connected under shared photoredox and sulfur chemistry assumptions [12]. Related work on prebiotic selection from complex mixtures shows that plausible chemistry can enrich biologically relevant building blocks without assuming a clean reagent list [6].

This strengthens the CDFD interpretation of autocatalytic closure. Closure is not merely a loop in an abstract reaction graph. It is a filter on messy chemical space:

$$\mathcal{N}_{t+1} = \mathcal{S}(\mathcal{N}_t, \Phi, C, \text{feedstock}, \text{cycle}), \quad (11)$$

where $\mathcal{S}$ selects reactions that persist through energy input, surface contact, hydration changes, and side-reaction pressure.

Status: Inferred

13 Nucleotide and Template Chemistry Boundary

Activated pyrimidine ribonucleotide synthesis under plausible conditions is a major anchor for any RNA-adjacent OOL story [14]. Mineral-surface oligomerization experiments provide a second anchor for polymer formation [4]. However, CDFD should distinguish three layers:

- precursor selection: making feedstock biased toward useful monomers;
- oligomer formation: building polymers longer than random hydrolysis permits;
- templating: copying sequence or pattern with bounded error.

Paper 7 reaches the first two layers only in principle. Paper 9 is where templating and error thresholds become explicit.

Status: Derived

14 Limitations and Open Problems

- No explicit racemization pathways or chiral mineral surfaces.
- Single scalar “pattern strength” hides sequence information.
- Environmental UV and radical chemistry not modeled.

Status: Open

15 Conclusion

This paper links flux-heavy environments to aromatic survivability and introduces chirality dynamics as a hypothesis. Paper 9 later formalizes replication fidelity and error thresholds [10, 3].

Status: Inferred

16 Reproducibility Note

`python supplementary_ool_07.py`

The script writes compact reproducibility outputs under `outputs/paper_07/`.

References

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